

PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION FOR
RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE-2019

SUBJECT: CHEMISTRY (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE: Attempt Five Questions in All. Attempt in Urdu or English.

Q No. 1: a) What is meant by corrosion? How it can be explained electrochemically? Also list methods to avoid them.

(10 Marks)

b) Define Gibbs free energy. How it can be related to equilibrium constant? Derive the relationship.

(10 Marks)

Q No. 2: a) Define the following: **(5X2=10 Marks)**

- i. Millimole ii. Equivalent iii. Tunneling effect
iv. Potential v. Entropy

b) Derive the equation of wave function for a particle moving in one dimension. **(10 Marks)**

Q No. 3: a) Differentiate between $K_4[Fe(CN)_6]$ and $[Fe(H_2O)_6]SO_4$ on the basis of CFT. Also explain the distortion present in copper complexes.

(4+6=10 Marks)

b) Write down postulates of VBT in case of transition metal compounds. Also comment, why ClO_4^- and H_2SO_4 has the same geometry,

(7+3=10 Marks)

Q No. 4: a) Define pollutant. How it produces adverse effect on living organisms and imbalance in the environment? **(10 Marks)**

b) Explain briefly the advantage and disadvantage of supplying sewage and sludge to the agriculture soil. **(10 Marks)**

Q No. 5: a) How will you prove " $H\psi = E\psi$ " by the Schrodinger wave equation? **(10 Marks)**

b) Briefly, explain the followings: **(5 + 5=10 Marks)**

- (i) Principal Quantum Number (ii) Azimuthal Quantum Number

Q No. 6: a) Define thermodynamics and enthalpy. Prove First law of thermodynamics by mathematical formulation. **(10 Marks)**

b) Explain the followings, **(5 + 5=10 Marks)**

(i) Gibbs free energy and its applications in different process

(ii) Change in enthalpy (ΔH) and relationship with change in energy (ΔE)

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- Q No. 7:**
- a)** Define hybridization and give examples. Write down the rules of hybridization and explain SP hybridization and SP^3 hybridization. **(10 Marks)**
 - b)** Describe the main postulates of Molecular Orbital Theory. Give its application in heteronuclear diatomic molecules with two examples. **(10 Marks)**
- Q No. 8:**
- a)** Discuss the nature of Alpha (α) rays and Gamma rays (γ). Describe the kinetic study of radioactive decay. Give example. **(10 Marks)**
 - b)** Define pollution, pollutants and contaminants. Describe the source, effects and control of carbon monoxide as pollutant. **(10 Marks)**



CSS VISTA
WITH SADIA ZAHOOR
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SUBJECT: CHEMISTRY (PAPER-II)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE: Attempt Five Questions in All. Attempt in Urdu or English.

- Q No. 1:** (i) Draw the structures of the following compounds: (5 Marks)
- | | |
|-----------------------|---------------------------------|
| a) Spiro [2.4]heptane | b) Bicyclo [2.2.2] cyclo octane |
| c) Hexane dioic acid | d) 4-oxobutanoic acid |
| e) Ethanoic anhydride | |

- (ii) Justify the following statement: (2x5=10 Marks)
- | | |
|--|------------------------------------|
| a) Dimethyl amine is stronger base than methyl amine which is stronger than ammonia | b) Phenol is stronger than ethanol |
| c) <i>m</i> – chloro-benzoic acid is a stronger acid than <i>p</i> – chloro-benzoic acid | |
| d) Pyrrole is a weaker base than Pyridine | |

- iii) Draw the resonance structures of the following compounds: (5 Marks)

- | | | |
|---|---|--|
| (a) $\text{CH}_2=\text{CH}-\text{CH}_3$ | (b) $\text{CH}_3=\text{CH}-\text{O}-\text{CH}_3$ | (c) $\text{CH}_3-\text{N}=\text{N}=\text{N}$ |
| (d) $\text{CH}_2=\text{CH}-\text{C}=\text{N}$ | (e) $\text{CH}_3-\text{C}-\text{CH}_2-\text{C}-\text{OC}_2\text{H}_5$ | |

- Q No. 2:** (i) Methanol is a good solvent for UV but not for IR Spectroscopy. Explain the fact. (5 Marks)
- (ii) Briefly describe the basic difference between Beer's and Lambert's laws of absorption of light. (5 Marks)
- (iii) Explain the working of double beam IR instrument with special discussion on the formation of a solution and recording of absorbance. (10 Marks)

- Q No. 3:** (a) Illustrate the difference between the following with examples: (3 + 5 + 4+5=17)
- | | |
|------|---|
| i. | Conjugation and Hyperconjugation. |
| ii. | Mesomeric Effect and Inductive Effect. |
| iii. | Chromophore and Auxochrome. |
| iv. | Electromeric Effect and Resonance Effect. |
- (b) Why esters have low boiling points than their corresponding acids? (3 Marks)

- Q No. 4:** (a) Explain the principle of UV and IR Spectroscopy. (3.5+3.5=7 Marks)
- (b) How would you distinguish between ethane ethylene and acetylene by IR spectroscopy? (6 Marks)
- (c) Differentiate between Bathochromic Shift and Hypsochromic Shift. (5 Marks)
- (d) What is the formula for absorbance of a substance in Electronic Spectroscopy? (2 Marks)

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- Q No. 5:** (i) Explain the steps involved in the phenomenon of "Walden Inversion". **(5 Marks)**
 (ii) Discuss the role of Nicol Prism for plane of polarized light. **(5 Marks)**
 (iii) Sketch the energy profile diagram and shapes of boat, twist boat, half chair and chair conformations of cyclohexane. **(10 Marks)**
- Q No. 6:** (i) Differentiate between the following terms with the help of appropriate examples.
 (a) Fats and oils (b) Soap and detergents (c) Lipids and wax
 (d) Iso-electronic point and flash- point. **(2x4=8 Marks)**
 (ii) Proteins exhibit a variety of coloured reactions, describe any four of these. **(8 Marks)**
 (iii) Hydrogen bonding plays an important role in protein structure, prove it. **(4 Marks)**
- Q No. 7:** (a) **What are Carbohydrates? Discuss their classification.** **(2+6=8 Marks)**
 (b) How do you differentiate between paper partition and adsorption paper chromatography? **(6 Marks)**
 (c) Define Detergent. What are its types and how it is different from soap? **(2+2+2=6 Marks)**
- Q No. 8:** (a) Explain the difference between chain growth and step growth polymerization? **(6 Marks)**
 (b) **Write a brief note on the following:** **(3.5x4=14 Marks)**
 i. TLC
 ii. Functions of Lipids
 iii. PTFE and its applications
 iv. P-type and N-type Semiconductors